Aegis Station

# Lunar Water Logistics

6 Million Tonnes of Lunar Water to Lunar Orbit, on a Fleet That Imports Nothing

Architecture Reference Document | Rev C — November 2026 | Not for Hardware Commitment

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## 1. Executive Summary

Aegis Station is a rotating habitat in a 100 km lunar orbit, shielded by three metres of water around the outer hull of each of its three rings. At the 1g baseline that shield is 6.0 million tonnes. Lifting that mass from Earth is not a plan; it is a number that ends the program. Every tonne of shield water is therefore mined at the lunar south pole, carried to the pad in a standard 45-tonne cartridge, and flown to orbit by an autonomous tanker.

Rev B made one structural change to the chain. The tanker flies on liquid oxygen and liquid hydrogen split from lunar water at the pad, from its first flight. Rev A baselined methalox with methane shipped from Earth and treated hydrolox as a later upgrade. At fleet cadence that methane was a recurring import of hundreds of tonnes a day landed on the Moon, a supply chain larger than every lunar landing to date, repeated indefinitely. Anything a program has to keep buying from Earth is the thing that ends it. The import is designed out before the first flight, and the honest cost moves from the launch manifest to a refinery-class power plant on the surface.

Rev C corrects the shield mass to the full 3 m shell of the station's geometry, 6.0 million tonnes where Rev B carried 3.3, and resizes the campaign to hold the fill time. The architecture is otherwise unchanged and is stated here as a baseline, not a closed design: 82 tankers, 3,690 tonnes a day, a fill of roughly 4.5 years, and the Lunar Orbital Propellant Depot as the fleet's first customer before the station exists. Section 11 lists what does not yet close.

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## 2. Water Requirement

The shield is a 3 m water layer on the outer hull of each ring, outboard of a 7 m utility band and a 40 m habitable depth. The requirement is set by geometry and the 1g rotation baseline, not by logistics.

| Parameter | Value |
|---|---|
| Shield thickness | 3 m of water, outer hull, all three rings |
| Shield radial span | 47 m to 50 m from the ring centreline, outboard of the utility band |
| Shield volume, three rings | ~6.0 million m³ |
| Water mass | ~6.0 million tonnes |
| Olympic pool equivalent | ~2,400 |
| Source | Lunar south pole, permanently shadowed craters. Earth-sourced water is excluded in every phase. |

Shield thickness is the multiplier on every number in this document: tankers, sorties, power, and years. The 3 m figure is held by the station design; the dose case for it is stated on the water-shield evaluation page, not here.

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## 3. Delivery Architecture

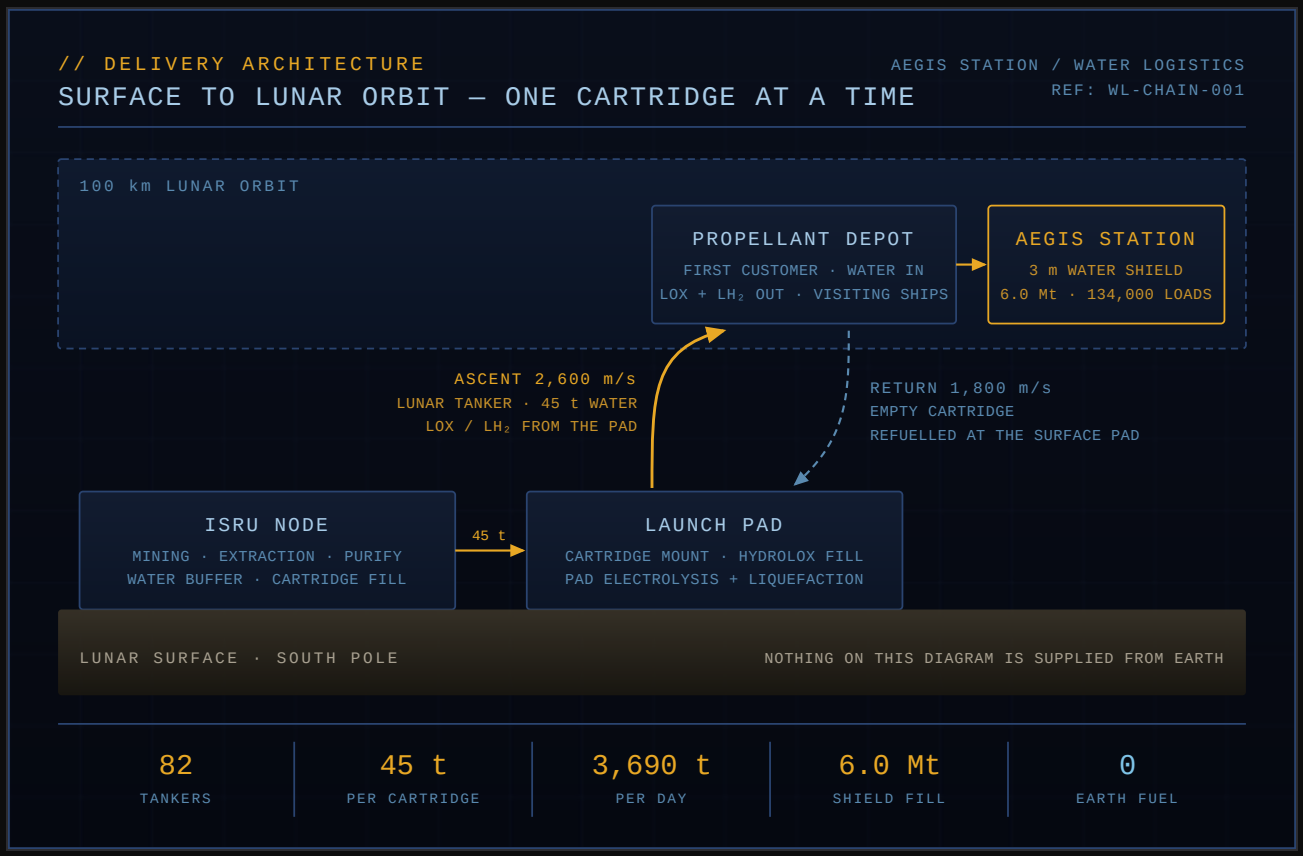



Fig. 1 — The delivery loop. Every sortie carries 45 t of water up and an empty cartridge down. Nothing on the loop is supplied from Earth.

| Parameter | Value |
|---|---|
| Fleet | 82 autonomous reusable tankers, VTOL, one ground console |
| Payload | 45 t of water per sortie, in one Modular Tank Cartridge |
| Ascent | 2,600 m/s to 100 km lunar orbit |
| Return | 1,800 m/s, empty cartridge, round-trip propellant loaded at the surface pad |
| Daily throughput | ~3,690 t/day |
| Fill time | ~4.5 years at that throughput |
| Sorties, total | ~134,000 |
| Propellant | LOX / $LH_2$, both split from lunar water at the pad |
| Earth-supplied consumables | None, in any phase |

### The Modular Tank Cartridge

The cartridge is the atomic unit of the chain: a 10 m × 2.5 m structural cylinder holding a flexible bladder of ~49 $m^3$, filled by vacuum draw at the ISRU node and emptied by inert-gas displacement at the destination. It is cargo the tanker carries, not part of the tanker, and it has one water port and one gas port that are identical at the node, the pad, the depot, and the station. Water is never a cryogen anywhere in the chain. It needs no insulation, no vacuum-jacketed lines, and no boil-off management, which is why it is the commodity that moves.

### Mission profile

0. **Depot first light.** A single tanker at low cadence lifts the first cartridges to the Lunar Orbital Propellant Depot and commissions end-to-end propellant production. Full fleet deployment is not required.
1. **Shield-fill campaign.** Sustained surface-to-orbit delivery to the station's shield reservoirs. This is the requirement that sizes the 82-vehicle fleet.
2. **Depot resupply.** Runs in parallel from first light and continues indefinitely after the fill. The depot is the fleet's first customer and a permanent one.
3. **Surface distribution.** Ballistic hops between ISRU and LUNET nodes at 160–940 m/s, redistributing water and LOX. Up to six hops per orbital propellant load.
4. **Rapid response.** One or two tankers held in standby; emergency water to a crewed outpost within 6–8 hours of dispatch.

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## 4. Propellant: Hydrolox From Day One

The Moon has water ice. Electrolysis splits it into liquid oxygen and liquid hydrogen, the two highest-performing chemical propellants there are, and the only ones the Moon can make from a single feedstock without importing anything. The tanker burns both from its first flight. There is no methane

phase, no transition, and no import to taper off later. The Short Hopper already flies LOX/$LH_2$ for the same reason; since Rev B every vehicle on the surface burns the same ISRU product.

| Parameter | Rev A (methalox) | Rev B (hydrolox) |
|---|---|---|
| Propellant | LOX / $CH_4$, methane imported from Earth | LOX / $LH_2$, both from lunar water |
| Specific impulse, vacuum | ~360 s | ~450 s |
| Launch mass | ~102 t | ~95 t |
| Ascent propellant | ~52 t | ~42 t |
| Round-trip propellant | ~57 t | ~45 t |
| Thrust, engines | ~165 kN, 3 × 60 kN | ~155 kN, 4 × 40 kN, BE-7 class, engine-out capable |
| Recurring Earth import per sortie | ~12.5 t methane | None |

Rev B propellant masses follow the rocket equation at the same payload, delta-v, and implied dry mass as Rev A. They are not a new sizing pass; see Section 11.

### The hydrogen objection

Liquid hydrogen at 20 K is hard to store and harder to move. This architecture answers that the same way LUNET does: *produce near, consume fast.* A tanker fills from a liquefier a few metres from the pad and is empty in orbit within hours. That is the shortest hydrogen dwell any lunar vehicle will have. The long-dwell problems, boil-off, thermal-cycle fatigue on transfer lines, and embrittlement, belong to depots holding $LH_2$ for months, not to a vehicle that burns its load the same day it is filled. Zero boil-off storage at 20 K has been demonstrated at twice the previous state of the art, and the industry's own next-generation lunar lander is hydrolox for precisely this reason: it is the propellant ISRU can actually make.

### The honest cost

Hydrogen is a ninth of water by mass, so a full tanker load means splitting several times its own hydrogen mass in water and liquefying the result. The tanker pad is therefore not a LUNET fill node, which is sized at tens to a few hundred kilowatts. It is a refinery-class facility with power to match, and it is the same plant the depot and the surface network already assume exists. That is one thing to build, once, on the Moon, rather than something to keep launching. Its size scales with sortie cadence and is an open item in Section 11.

> **Design rule.** No vehicle in the surface fleet may depend on a consumable that must be resupplied from Earth. *Methane failed this rule. Hydrolox passes it because the feedstock is the cargo.*

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# 5. Refuelling: Method and Location

The loading-site capacity row in Section 11 makes pad turnaround the number the fleet keys off. Two decisions set it: how propellant gets aboard, and where. This section closes both under the fleet-analysis assumptions of Section 11: a full cartridge is at the pad, an empty is at the station, and propellant is available when called for.

### What has to be moved each cycle

| Item | Value |
|---|---|
| Round-trip propellant | ~45 t: ~6.4 t $LH_2$ at 20 K, ~38.6 t LOX at 90 K |
| Tank volume | ~90 $m^3$ $LH_2$ plus ~34 $m^3$ LOX, about 125 $m^3$, two and a half times the water cartridge |
| Fill rate, fluid transfer | Shuttle-era ground systems loaded $LH_2$ at ~10 kg/s. At 2–5 kg/s $LH_2$ and ~10 kg/s LOX, in parallel, a full load takes 60–75 minutes |
| Cryogenic connections per cycle | Two, either way. A fluid coupling connects a hose; a swappable module connects the module's feed to the engines. Neither option removes the connect cycle; they move it. |

### Method: fluid couplings versus swappable propellant modules

| Turnaround step | Fluid couplings | Swappable modules |
|---|---|---|
| Land, safe, vent checks | 15 min | 15 min |
| Cartridge swap: unlatch empty, mount full, latch, check | 50 min, in parallel with fill | 50 min, in parallel with module swap only if two heavy handlers work one vehicle at once |
| Propellant aboard | Couple and chill lines 20 min, transfer 60–75 min, disconnect, leak check and purge 15 min | Disconnect and remove spent module 25 min, install, latch, connect and leak-check 35 min, chill engine feed 10 min |
| Inspection and prelaunch | 30 min | 30 min |
| **Pad time, steps overlapped** | **~2.6 h** | **~2.0 h** |
| Pad time, sequential | ~3.5 h | ~2.7 h |

| Consequence | Fluid couplings | Swappable modules |
|---|---|---|
| Pads for 82 sorties/day | 9 at 2.6 h; 11 hold the cadence at the 3.2-hour ceiling | 7 at 2.0 h; 11 allow up to 3.2 h |
| Vehicle dry mass | Tanks are primary structure, as on every lander flown | Tanks become cargo, so the airframe needs its own thrust structure plus a flight-rated structural, fluid and electrical interface: about +2–3 t |

| Consequence | Fluid couplings | Swappable modules |
|---|---|---|
| Payload per sortie | 45 t | ~42–43 t at the same launch mass, roughly 5 % more sorties over the campaign, so 5 % more engines, propellant, pad-hours and years |
| Where the cycle wear lives | Ground-side hose couplings: cheap, replaceable on a schedule, no flight mass. This is the LUNET rule for cryogenic lines applied to the pad. | A flight-loaded interface on the vehicle that must survive ~1,600 connect cycles and every launch and landing between them |
| Ground hardware | Buffered LOX and $LH_2$ store, transfer lines, one set of couplings per pad | The same store and lines, plus a fill bay, a module handling system for a ~50 t cryogenic unit, and a pool of ~45 modules at a few tonnes each |
| Boil-off | Fill on demand; losses are line chilldown per cycle | Filled modules wait on the apron and lose $LH_2$ while they do |
| Off-nominal | A failed coupling closes one pad | A failed module is one module; a failed interface grounds the vehicle |

The module buys about 0.6 hours of pad time per sortie, which at fleet cadence is about two pads. Pads are among the cheapest things on the site. It pays for that with two to three tonnes of dry mass on every flight for the whole campaign, a heavy cryogenic handling system that the water cartridge does not need because water is not cryogenic, and a flight-rated interface that carries the same connect-cycle count as the hose coupling it replaces but under launch loads. The water cartridge is swappable because water tolerates it. Propellant does not get the same treatment for the same reason the network moves water and not hydrogen.

### Location: surface only versus depot top-off

| Consideration | Surface only | Surface plus depot top-off for the return leg |
|---|---|---|
| Return-leg propellant | Lifted from the surface, ~3–5 t | Taken on at the depot, ~3–5 t of hydrolox per sortie |
| Ascent saving | None | Not lifting the return propellant saves ~6–9 t of ascent propellant, worth roughly +3–4 t of payload per sortie |
| What it costs in orbit | Nothing | 3–5 t of hydrolox per sortie is ~6 t of water electrolysed at the depot per sortie, ~530 t/day at fleet cadence, of order 160 MW of orbital power and a large 20 K storage bank |
| Berth time | Cartridge swap only | Cartridge swap plus a second cryogenic connection and transfer per cycle, on the resource the capacity row says is as tight as the pad |

| Consideration | Surface only | Surface plus depot top-off for the return leg |
|---|---|---|
| Where the power already is | The refinery is on the surface and is sized for the ascent load anyway; the return leg is a ~10 % increment on it | Moves a tenth of the refinery into orbit for a tenth more payload |

Depot top-off is the right answer for the Hauler, which cannot lift its return propellant from Earth economically, and for any vehicle that arrives from elsewhere. For a tanker that departs from a pad with a refinery on it, the return leg is a small increment on a plant that already exists, and buying it in orbit instead costs berth time and an orbital power plant. The tanker fills at the surface. Depot top-off stays available as an option for contingency and for post-fill operations when the depot has surplus, and the depot interface is unchanged.

> **Decision.** Fluid couplings at the surface pad, filled on demand from a buffered store, cartridge swap in parallel with propellant transfer, ground couplings treated as scheduled consumables. Turnaround target 2.6 hours, ceiling 3.2. *Swappable propellant modules and routine depot top-off are rejected for the water tanker. Fifteen pads are recommended over eleven for margin: eleven hold the cadence only at the 3.2-hour ceiling, fifteen hold it with a pad occupancy up to 4.4 hours, so a slow transfer or a pad out of service does not stall the fill.*

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## 6. The Depot as First Customer

Every lunar-water architecture faces the same deadlock: surface extraction is hard to justify without an orbital customer, and orbital hardware is hard to justify without a proven resource. The Lunar Orbital Propellant Depot breaks it by being the earliest useful thing in lunar orbit. Its hardware can be emplaced and stand dormant; its commissioning date is set by the first cartridge to arrive, and its existence is what makes that date matter.

- **Water in, propellant out.** Depot electrolysis produces LOX and $LH_2$ from cartridge water. With a hydrolox fleet the hydrogen is a product, not a vented byproduct: the depot holds both propellants for visiting vehicles and for contingency top-off of the tanker fleet. Routine tanker refuelling is at the surface pad, per the decision above, so the depot is not in the daily cycle.
- **Methane for the Hauler only.** The Earth–Aegis Long-Hauler still burns methalox for its return; it brings its own methane as manifest cargo. That import is declared, bounded, and never touches the tanker fleet.
- **Standard interfaces.** The cartridge and its two ports are the same at the depot as at the pad. A depot that only its builder can dock with is a fuel tank, not infrastructure.
- **Beachhead.** The depot's pressurised node is the program's first shirt-sleeve presence in lunar orbit, before any ring is assembled.

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# 7. ISRU and Extraction

Ice-bearing regolith is excavated from permanently shadowed craters near the south pole and heated by microwave or radiant thermal rigs; the ice sublimes, is cold-trapped, condensed, and filtered to a single water grade that serves both cargo and propellant. An ambient surface buffer absorbs production swings and feeds two consumers: cartridge fill for the tanker, and pad-side electrolysis and liquefaction for the tanker's own propellant.

- Deposit assessment precedes extraction; the subsurface ice assessment capability and the WOK prospecting rover are the instruments aimed at that gate.
- Mining, extraction, and transfer are autonomous and modular; power is polar solar with fission for the lunar night.
- For every tonne delivered to orbit, roughly another tonne and a quarter is split for propellant. The mining rate, not the delivery rate, sizes the node.

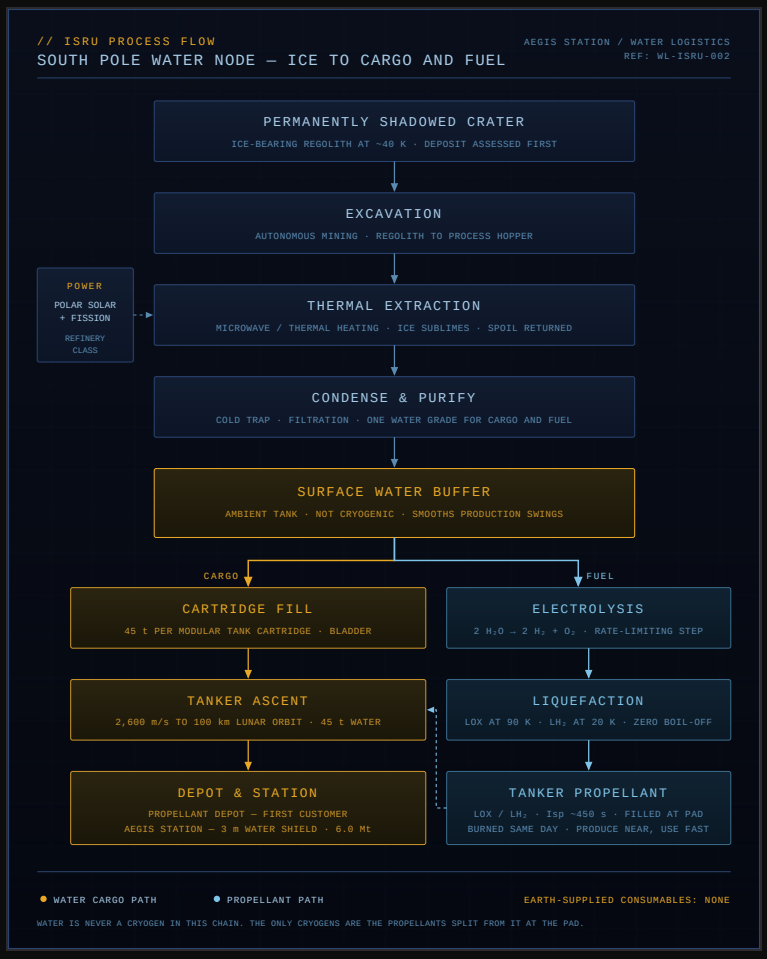



Fig. 2 — From ice in the crater to water in a cartridge and hydrolox in the tanker. The buffer is ambient; the only cryogens are the propellants split from it.

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# 8. Logistics Infrastructure

### Surface

- Launch pads with dust control and blast deflection
- Cartridge fill, staging, and ground inventory of displaced inert gas

- Pad-side electrolysis and liquefaction: the refinery that fuels the fleet
- LUNET utility nodes for surface mobility, refuelling, and inter-node exchange
- Fleet Commander ground console: dispatch, pad deconfliction, berth queuing, standby reserve

### Orbital

- Lunar Orbital Propellant Depot: cartridge reception, electrolysis, LOX and $LH_2$ storage, tanker top-off
- Station cartridge docking ports and water transfer lines to the shield reservoirs
- Autonomous transfer monitoring, inventory reconciliation, and fault management

### Shield fill

Rings receive shield water from the start of assembly through evenly spaced fill ports. Fill sequencing preserves mass balance across each ring, and spin-up to 1g follows structural and systems verification of the complete ring assembly, so the shield is filled and balanced before the ring rotates.

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## 9. Quantities

Cost is not yet estimated. This revision states what the fill moves and what moves it, which are the numbers a cost would be built from.

| Item | Quantity |
|---|---|
| Shield water delivered | ~6.0 million tonnes |
| Cartridge loads | ~134,000 |
| Water mined, cargo plus propellant | ~14 million tonnes; ~8,300 t/day at fleet cadence |
| Tankers | 82, each flying about 1,600 cycles |
| Pads and berths | 11 at the 3.2-hour ceiling; 15 recommended |
| Engines over the campaign | ~5,400–10,700; 800–1,600 t landed on the Moon |
| Surface refinery | Of order 1.5 GW continuous |
| Cartridge pool, mining, and depot hardware | Not yet sized |

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## 10. Long-Term Applications and Strategic Value

The extraction and delivery chain does not retire when the shield is full. The same fleet, pad, and depot serve depot resupply, surface distribution, emergency response, and propellant staging for anything leaving cislunar space, still without an Earth-supplied consumable.

- LOX and $LH_2$ production for cislunar and deep-space vehicles

- Depot refuelling in lunar orbit and at Earth–Moon staging points
- Life support reserves and shielding water for future habitats and platforms
- Water as a priced, deliverable orbital commodity with published interfaces
- A replicable model for any station built where water is available

*The water economy doesn't end with Aegis. It begins with it.*

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# 11. Open Items

Rev B fixed the propellant decision. Rev C resizes the campaign for the corrected shield mass and leaves the vehicle cycle as stated. The following do not yet close and are carried as open, not hidden.

### Scope and assumptions for the fleet analysis

The tanker fleet is analysed as a vehicle cycle between a surface pad and a station berth. Three things are assumed rather than analysed here, because each belongs to another part of the program:

A1 **A full cartridge is available at the pad whenever a tanker is ready to lift.** Mining, extraction, cartridge fill, and the size of the cartridge pool are upstream of the fleet and sized separately.

A2 **An empty cartridge is available at the station whenever a tanker berths.** Shield-side discharge and cartridge handling are station systems.

A3 **Refuelling is timely wherever it happens, surface or station.** Refinery and depot capacity are their own items; they set the ceiling on cadence but are not part of the vehicle cycle.

Under those assumptions the fleet's own limits are pad-hours, berth-hours, orbital phasing, airframe availability, and turnaround mechanics. Where refuelling happens and how propellant gets aboard are decided in Section 5; the turnaround target that follows from it is carried below.

| Item | Status |
|---|---|
| Daily turnaround | 3,690 t/day at 45 t per sortie is one flight per tanker per day for the whole campaign, about 1,600 cycles per airframe. This is a requirement, not an assumption; the fill time is derived from it. It is an airliner cadence in an environment gentler than any reusable booster faces: no reentry, no aerodynamic load, one-sixth-g landings. What it demands is stated in the rows below: engines as scheduled replacement items, a pad turnaround of 2.6 hours, and a refinery sized to the cadence. No reusable orbital-class vehicle has yet flown daily; that is what the fleet has to demonstrate, not a reason to plan for less. |
| Engine pipeline | Each engine position sees roughly 500 hours of burn over an airframe life, many times any demonstrated engine life, so engines are scheduled replacement items. At 50–100 sorties per engine set, four engines per vehicle, the campaign consumes about 5,400–10,700 engines, three to seven a day from a single printed part number, 800–1,600 t landed on the Moon. Engine life is the lever: 200 sorties per set quarters the count. Life-expired engines stay at the pad as the first refurbishment demand on the surface. |
| Vehicle dry mass | The vehicle must close at 45 t of payload to lunar orbit. The stated launch and propellant figures imply roughly 5 t of structure, tanks, engines, and cartridge shell, which is aggressive for a lander. A bottom-up mass budget that closes at 45 t, with the launch mass that requires, is the open item. Not done in this revision. |

| Item | Status |
|---|---|
| Surface power | Electrolysis and hydrogen liquefaction at fleet cadence are of order 1.5 GW continuous, about 18 MW per daily sortie. LUNET node sizing does not cover this; the pad refinery is a separate facility and is not yet sized. |
| Cartridge reuse | The cartridge is infrastructure, not a consumable, but its cycle life and bladder material are undefined. Fleet size of cartridges bounds throughput. |
| Loading-site capacity | Eighty-two sorties a day is one launch every 18 minutes. With eleven pads and eleven berths, each tanker may hold a pad for at most 3.2 hours and a berth for at most 3.2 hours, all steps included. The fleet dispatch ADR carries a 4–8 hour placeholder for the swap; this requirement supersedes it. Pads are cheap against megawatts: seventeen pads relax the turnaround to five hours. A tanker waiting on propellant or a cartridge must not hold a pad; it waits on an apron, so pad occupancy measures servicing only. |
| Turnaround target | Closed to a target by Section 5: fluid couplings at the surface, 2.6 hours with cartridge swap and propellant transfer overlapped, 3.2 hour ceiling for eleven pads. Still open: whether a 2.6 hour turnaround is achievable in practice with the cargo and coupling hardware as designed. Fifteen pads recommended for margin. |
| Airframe availability | Eighty-two sorties a day requires eighty-two airframes available every day. Airframes in engine swap or inspection are in addition to that number, not subtracted from it. The size of the maintenance rotation follows from engine-swap time and inspection intervals and is not yet set. |
| Resource base | Shield water plus propellant water is about 14 million tonnes against polar ice estimates that span two orders of magnitude. Deposit assessment is the gate, and it is why the program is waiting. |

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# 12. Changes From the Previous Revision

### Rev B to Rev C

| Item | Rev B | Rev C |
|---|---|---|
| Shield water | ~3.3 Mt | ~6.0 Mt, the full 3 m shell between 47 m and 50 m on the 350 m ring centreline |
| Olympic pool equivalent | ~1,320 | ~2,400 |
| Fleet | 45 tankers | 82 tankers, sized to hold the ~4.5-year fill |
| Daily throughput | ~2,025 t/day | ~3,690 t/day |
| Sorties, total | ~73,333 | ~134,000 |
| Launch cadence | One every 32 minutes | One every 18 minutes |
| Pads and berths | Six at the 3.2-hour ceiling, eight recommended | Eleven at the 3.2-hour ceiling, fifteen recommended |
| Engines over the campaign | ~3,000–6,000 | ~5,400–10,700 |
| Surface refinery power | Hundreds of megawatts | Of order 1.5 GW |
| Cost figures | Per-kilogram rates and campaign totals | Removed; Section 9 states quantities |

### Rev A to Rev B

| Item | Rev A | Rev B |
|---|---|---|
| Tanker propellant | LOX/$CH_4$, methane imported from Earth, hydrolox as a later phase | LOX/$LH_2$ from lunar water, first flight onward |
| Earth-supplied consumables | ~12.5 t methane per sortie for the first years | None |
| Water handling | "Cryogenic transport tanks", "cryogenic storage tanks", "cryogenic transfer lines" | Ambient water in bladder cartridges; only the propellants are cryogenic |
| Surface nodes | Remote Operations Nodes | LUNET utility nodes plus a refinery-class pad |
| Depot | Not described | First customer; commissions on first cartridge; hydrogen retained for tanker top-off |
| Open items | Not stated | Section 11 |
| Refuelling | Not stated | Fluid couplings at the surface pad, fill-on-demand from a buffered store; swappable propellant modules and routine depot top-off rejected (Section 5) |

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